

descSPIM-SLIM: Python Control Installation Instructions

These instructions outline the procedure for controlling the descSPIM-SLIM using Python to perform 3D multicolor imaging.

※This guide is intended for use with Windows 11.

※Computer specs require at least 64 GB of RAM.

1. Environment creation

1-1. Application Installation

1-1-1. Installation of Python

1-1-2. Installation of Anaconda

1-1-3. Installation of ThorCam™ software

1-1-5. Installation of Kinesis® software

1-2. Environment Setup

1-2-1. Creation of a new environment

1-2-2. Installation of required packages

2. Execute Python code

2-1. Homing Cell (Connection and Initialization)

2-2. ThorCam Configuration

2-3. Velocity Setting GUI Cell

2-4. Focusing Procedure

2-5 Synchronized Imaging

3. Troubleshooting

1. Environment creation

1-1. Application Installation

Install the required environment from the following URLs.

1-1-1. Installation of Python

<https://www.python.org/>

Download the installer and follow the instructions to install Python. Select "Add Python to PATH" during installation and add Python to the PATH environment variable.

Currently, this PC is using Python 3.13.5. Different versions and bit numbers may lead to errors.

1-1-2. Installation of Anaconda

<https://www.anaconda.com/download>

Run the downloaded installer. Follow the instructions provided by the installer. Usually, the default settings are fine. If you select the option "Add Anaconda to my PATH environment variable", you can use Anaconda from the command line (Windows).

We recommend downloading the latest version of Anaconda Distribution. Currently on this PC, Anaconda CLI, version 0.5.2 is used. Search and open Anaconda prompt from home.

```
jupyter notebook
```

Execute the above command and jupyter notebook should be up and running.

1-1-4. Installation of ThorCam (application for Thorlabs cameras)

https://www.thorlabs.co.jp/software_pages/ViewSoftwarePage.cfm?Code=ThorCam

Install ThorCam Software for 64-Bit Windows from the above URL. Follow the installation guide. Use version 3.7.0.6.

1-1-5. Installation of Kinesis® software (Thorlabs motor application)

https://www.thorlabs.co.jp/newgrouppage9.cfm?objectgroup_id=10285

Install Kinesis® software from the above URL. Follow the installation guide. Use version 1.14.57

1-2. Environment Setting

1-2-1. Creating a new environment

Launch Anaconda prompt and execute the following commands, this will create and activate a new environment (myenv).

```
python -m venv myenv
myenv\Scripts\activate
```

Confirm that the environment is activated. You should see the following in your terminal window:

```
(myenv) C:\Users\username>
```

1-2-2. Install the required packages

To run the provided Jupyter Notebook (251108_success_descSPIM-SLIM_Python_completion_file.ipynb), install the following packages.

```
pip install jupyter
# Required to open and run the .ipynb notebook file.

pip install pythonnet
# Required to import 'clr' and control the Thorlabs Kinesis hardware.
```

2. Execute Python code

2-1. Homing Cell (Connection and Initialization)

Before running the “Homing” cell, first check and edit the following lines in the same cell:

```
SAMPLE_SERIAL = "26006224" # sample stage
CAMERA_SERIAL = "26006304" # camera stage
```

These are the serial numbers of the KST201 controllers. They differ for each individual device. Please replace them with the serial numbers printed on your own controller units.

Then, run the “Homing” cell. This cell both confirms your device connections and performs initial homing for the sample and camera stages.

When executed, it will:

1. Load the Kinesis DLLs and initialize the environment
2. Connect to both KCube controllers
3. Start polling and enable motion control
4. Perform homing (return-to-origin) for both stages.

Wait until [OK] and [Homing complete] appear in the console — this confirms that both devices are recognized and ready.

2-2. ThorCam Configuration

Follow the “ThorCam Configuration” cell for setup instructions:

- Exposure Time → 300 ms
- Frames per Trigger → Continuous
- Hardware Trigger → Mode: None, Polarity: On High

Close the *Settings* window after applying the changes. No code execution is required.

2-3. Velocity Setting GUI Cell

Run the “Velocity Setting GUI” cell. A small window will appear to input the velocity and acceleration for both stages.

Recommended parameters:

- Sample → MaxVelocity = 0.100 mm/s, Acceleration = 2
- Camera → MaxVelocity = 0.0342 mm/s, Acceleration = 2

Click “Set Speeds” to connect, apply the parameters, and automatically disconnect.

2-4. Focusing Procedure

Using the velocity parameters defined in Steps 2-2 and 2-3, adjust the sample position to locate the plane closest to the objective lens where the image first comes into focus. Once this focal plane is identified, move both the sample and camera stages simultaneously toward the near side (closer to the objective) until the sample image just disappears from view.

Stop the stages immediately at this point and define that position as the imaging start position.

It is recommended to halt movement as soon as the sample is no longer visible to ensure precise and repeatable positioning.

2-5 Synchronized Imaging

Run the “Synchronized Imaging” cell to start synchronized movement.

- Sample → 13 mm @ 0.020 mm/s

- Camera → 0.00684 mm/s (refractive-index-corrected speed)

When the console displays:

```
[POS] start: sample=xxx mm, camera=yyy mm,
```

immediately start ThorCam recording to achieve synchronized imaging.

When operating ThorCam, do not change any of the settings configured in Step 2.

Press Rec, select 8-bit TIFF (without annotation), specify any desired file name and save destination, and start recording. Starting the recording in ThorCam immediately sends a trigger signal from the SMA connector to the sample stage, initiating stage movement without minimal delay.

After imaging is complete, execute the next cell to return both stages to the imaging start position.

This allows you to change the excitation wavelength and perform repeated imaging, enabling multicolor 3D reconstruction.

It is recommended to record the starting positions of both stages in advance as a precaution.

3. Troubleshooting

Below are common issues that may occur during stage operation, along with their possible causes and recommended solutions.

☐ **Problem 1: The stages do not move when running the “Homing” cell**

Possible Cause:

Connection failure between the controller and the stage, or between the controller and the computer.

Solution:

- Verify that the controller is properly connected to the stage using the supplied cable.
- Check that the USB cable from the controller is securely connected to your laptop.
- Opening Thorlabs Kinesis® software can be useful to confirm that the device is recognized.

If the device does not appear there, a connection issue is the most likely cause.

☐ **Problem 2: The stages do not move even though the connections are correct**

Possible Cause:

Incorrect serial numbers assigned in the code.

Solution:

Ensure that the serial numbers specified for

```
SAMPLE_SERIAL = "xxxxxx"  
CAMERA_SERIAL = "xxxxxx"
```

- match exactly the serial numbers printed on your own KST201 controller units.
- The devices will not respond if incorrect serial numbers are entered.

☐ **Problem 3: The stage stops in the middle of imaging**

Possible Cause:

The travel range of the KCube stepper stage is limited, and the imaging start position was set too far along the range.

Explanation:

- The working distance of the stage is 25 mm. For a cleared E12.5 mouse embryo with a crown-rump length of approximately 13 mm, imaging typically requires about 13 mm of sample stage travel and $13 \text{ mm} \times 0.342 = 4.446 \text{ mm}$ of camera stage travel. If the imaging start position is set too close to the stage's end of travel, the stage may reach its limit and stop during imaging.

Solution:

- Always define the imaging start position with a sufficient safety margin.
For example, the sample stage should begin at least several millimeters before the 13 mm travel point (e.g., below ~12 mm) to ensure continuous motion.

☐ **Problem 4: The stage does not move after a previous run was interrupted**

Possible Cause:

If a previous cell stopped unexpectedly, the connection may still remain active in the background, preventing the next cell from running properly.

Solution:

- Normally, each cell automatically connects and disconnects the devices independently. However, if a cell was interrupted or an error occurred, a background

connection may persist. In this case, restart the Jupyter kernel to reset the environment and ensure proper reconnection before re-running the imaging cell.